

Technical University of Denmark

Written examination: 14th December 2023, 9 AM — 1 PM.

Course name: Introduction to Machine Learning and Data Mining.

Course number: 02450.

Aids allowed: All aids permitted.

Exam duration: 4 hours.

Weighting: The individual questions are weighted equally.

The exam is multiple choice. All questions have four possible answers marked by the letters A, B, C, and D as well as the answer “Don’t know” marked by the letter E. Correct answer gives 3 points, wrong answer gives –1 point, and “Don’t know” (E) gives 0 points.

This exam only allows for electronic hand-in.

You hand in your answers at <https://eksamen.dtu.dk/>. To hand in your answers, write them in the file `answers.txt` (this file is available from the same place you downloaded this file). When you are done, upload the `answers.txt` file (and nothing else). Double-check that you uploaded the correct version of the file from your computer.

Do not change the format of `answers.txt`

The file is automatically parsed after hand-in. Do not change the file format of `answers.txt` to any other format such as `rtf`, `docx`, or `pdf`. Do not change the file structure. Only edit the portions of the file indicated by question marks.

Answers:

1	2	3	4	5	6	7	8	9	10
C	B	A	D	A	D	C	A	B	A
11	12	13	14	15	16	17	18	19	20
A	B	C	B	C	C	A	B	D	D
21	22	23	24	25	26	27			
B	C	D	D	B	C	A			

	Attribute description	Abbrev.
x_1	Temperature measured in Celsius ($\mathbb{R}_{\geq 0}$)	Temperature (T)
x_2	Ambient pressure measured in millibar ($\mathbb{R}_{\geq 0}$)	Pressure (AP)
x_3	Relative humidity measured in percentage ($\mathbb{R}_{\geq 0}$)	Humidity (RH)
x_4	Exhaust vacuum measured in cm of mercury ($\mathbb{R}_{\geq 0}$)	Vacuum (EV)
y	Net hourly electrical energy output level	

Table 1: Description of the attributes in the Combined Cycle Power Plant (CCPP) dataset used in this exam. The dataset is used for determining the net hourly electrical energy output level. The dataset consists of $N = 9568$ observations and we consider the two classes $y = 1$ and $y = 2$ (corresponding to Low and High energy production, respectively).

Question 1. We consider the four attributes of the CCPP dataset¹ in Table 1. A scatter plot matrix for these attributes is shown in Figure 1.

Which one of the following matrices represents the empirical correlation matrix for these attributes?

A. $\hat{C} \approx \begin{bmatrix} 1.00 & 0.85 & -0.51 & -0.55 \\ 0.85 & 1.00 & -0.42 & -0.33 \\ -0.51 & -0.42 & 1.00 & 0.72 \\ -0.55 & -0.33 & 0.72 & 1.00 \end{bmatrix}$

B. $\hat{C} \approx \begin{bmatrix} 1.00 & 0.85 & -0.51 & 0.55 \\ 0.85 & 1.00 & -0.42 & -0.33 \\ -0.51 & -0.42 & 1.00 & 0.09 \\ 0.55 & -0.33 & 0.09 & 1.00 \end{bmatrix}$

C. $\hat{C} \approx \begin{bmatrix} 1.00 & 0.85 & -0.51 & -0.55 \\ 0.85 & 1.00 & -0.42 & -0.33 \\ -0.51 & -0.42 & 1.00 & 0.09 \\ -0.55 & -0.33 & 0.09 & 1.00 \end{bmatrix}$

D. $\hat{C} \approx \begin{bmatrix} 1.00 & 0.25 & -0.51 & -0.55 \\ 0.25 & 1.00 & -0.42 & -0.33 \\ -0.51 & -0.42 & 1.00 & 0.09 \\ -0.55 & -0.33 & 0.09 & 1.00 \end{bmatrix}$

E. Don't know.

Solution 1.

- The correlation of attributes $\text{corr}(x_3, x_4) = 0.72$ is unrealistic as it seems that their true correlation is ≈ 0 .
- The scatter plot shows that the correlation of attributes $\text{corr}(x_1, x_4) < 0$.

¹Dataset obtained from <https://archive.ics.uci.edu/dataset/294/combined+cycle+power+plant>

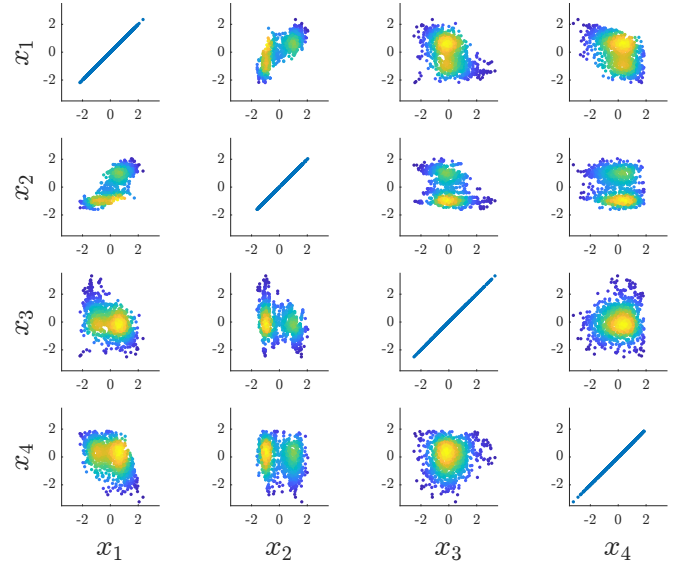


Figure 1: The scatter plot matrix for the four attributes x_1, x_2, x_3, x_4 of the CCPP dataset described in Table 1. In the off-diagonal panels, the color represents the associated density.

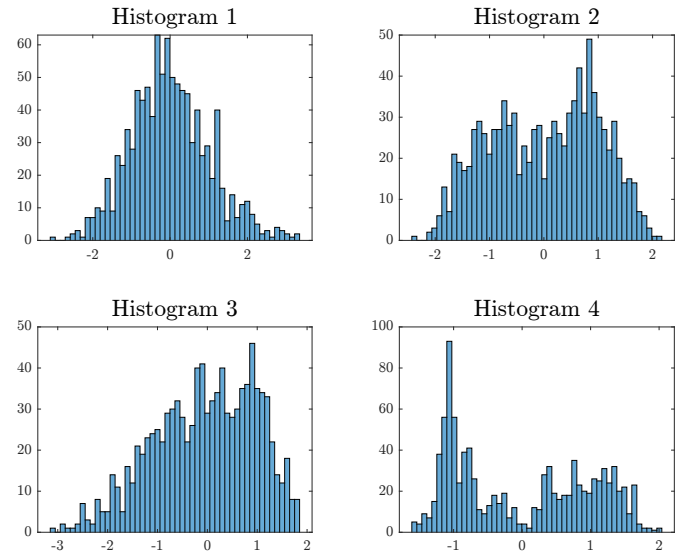


Figure 2: Histograms of the attributes in which their order has been permuted.

- This is the correct solution.
- The correlation of attributes $\text{corr}(x_1, x_2) = 0.25$ seems to be unrealistic as the attributes are strongly correlated.

Question 2. We consider the histograms of the four attributes in Figure 2. Match the histograms to the attributes by using the information regarding the distribution of the points from Figure 1.

Which one of the following statements is *correct*?

- A. Histogram 1: x_3 , Histogram 2: x_4 ,
Histogram 3: x_1 , Histogram 4: x_2 .
- B. Histogram 1: x_3 , Histogram 2: x_1 ,
Histogram 3: x_4 , Histogram 4: x_2 .**
- C. Histogram 1: x_2 , Histogram 2: x_3 ,
Histogram 3: x_1 , Histogram 4: x_4 .
- D. Histogram 1: x_2 , Histogram 2: x_1 ,
Histogram 3: x_3 , Histogram 4: x_2 .
- E. Don't know.

Solution 2. We can easily see that x_2 corresponds to Histogram 4 due to the two clearly separated modes. Then, x_1 corresponds to Histogram 2, as it has to modes but not clearly separated. The Histogram 3 corresponds to x_4 as the mode is above 0, while Histogram 1 corresponds to x_3 as the mode is below 0.

Question 3. We consider a subset of $N=5$ points from the CCPP dataset $\mathbf{X}$ where each row corresponds to an observation. We apply Principal Component Analysis (PCA) using the Singular Value Decomposition (SVD) and the resulting matrices $\mathbf{\Sigma}$, $\mathbf{V}$ are given below. The observation $\mathbf{x}_4 = [1.4 \ 1 \ 0.1 \ -2.8]^T$ is projected and then reconstructed using the first two principal components and the mean. The data matrix is

$$\mathbf{X} = \begin{bmatrix} -0.6 & -0.6 & 2.5 & -0.1 \\ -0.8 & -0.3 & -1 & 1.2 \\ -0.7 & 0.3 & -0.2 & -0.1 \\ 1.4 & 1 & 0.1 & -2.8 \\ -0.2 & 0.8 & -1.2 & 0.7 \end{bmatrix},$$

with $\tilde{X}_{ij} = X_{ij} - \frac{1}{N} \sum_{n=1}^N X_{nj}$ the centered matrix and the singular value decomposition

$$\tilde{\mathbf{X}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T, \quad \mathbf{\Sigma} \approx \begin{bmatrix} 3.7 & 0 & 0 & 0 \\ 0 & 3.04 & 0 & 0 \\ 0 & 0 & 0.56 & 0 \\ 0 & 0 & 0 & 0.48 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{V} \approx \begin{bmatrix} 0.43 & -0.26 & 0.22 & -0.84 \\ 0.17 & -0.37 & 0.81 & 0.42 \\ 0.33 & 0.88 & 0.34 & -0.01 \\ -0.82 & 0.14 & 0.42 & -0.36 \end{bmatrix}$$

Which one of the following reconstructions is *correct*?

- A. $\hat{x}_4 \approx [1.35 \ 1.11 \ 0.13 \ -2.77]^T$
- B. $\hat{x}_4 \approx [1.42 \ 1.15 \ 0.15 \ -2.91]^T$
- C. $\hat{x}_4 \approx [1.53 \ 0.97 \ 0.19 \ -2.35]^T$
- D. $\hat{x}_4 \approx [1.6 \ 0.91 \ 0.11 \ -2.69]^T$
- E. Don't know.

Solution 3. First we need to compute the mean of the data μ and the projection subspace using only the first two columns $\hat{V} = V_{1:2}$.

1. This is the correct answer. We need to apply for the reconstruction: $\hat{x} = \mu + \hat{V}\hat{V}^T(x - \mu)$
2. This is wrong, $\hat{x} = \mu + \hat{V}\hat{V}^T x$
3. This is wrong, $\hat{x} = \hat{V}\hat{V}^T(x - \mu) + [0, 0.1, 0.1, 0.2]^T$
4. This is wrong, $\hat{x} = \hat{V}\hat{V}^T x$

Question 4. We consider again the PCA result of the CCP dataset.

Which one of the following statements is *correct*?

- A. The first two eigenvectors explain less than 90 of the total variance.
- B. With respect to the first eigenvector, low values of Temperature, Pressure, Humidity and a high value of Vacuum results in a positive projection.
- C. For every dataset, if we do not use all the principal components, then we are not able to reconstruct perfectly the given data.
- D. For every dataset it holds for the eigenvectors computed by SVD that the angle between every pair of distinct eigenvectors is 90° degrees and the Euclidean norm of every individual eigenvector is equal to one.**
- E. Don't know.

Solution 4.

- This is wrong, since using the formula $(\sigma_1^2 + \sigma_2^2) / \sum_{i=1}^4 \sigma_i^2 \approx 0.98$.

- This is wrong, as the attributes lie in the interval $[0, +\infty)$, so the forth attribute due to $V_{4,1} = -0.88$ will push the projection to the negative values.
- This is wrong, as there might exist a dataset which perfectly lies on a linear subspace of dimension d within the high dimensional space of dimension D . In this case, d eigenvectors are enough to reconstruct perfectly the data.
- This is true, as the eigenspace is by definition always orthonormal.

Question 5. Which one of the following statements regarding logistic regression is *correct*?

- A. Consider a regularized logistic regression model for a binary classification problem. Assume that during training we over-regularize ($\lambda \rightarrow +\infty$) the weights while the intercept is not regularized. The resulting model will classify every test point to the class with the highest number of training points.**
- B. We can find the optimal weights for a regularized logistic regression classifier using the formula $w^* = (\tilde{X}^T \tilde{X} + \lambda I)^{-1} \tilde{X}^T y$.
- C. The logistic regression classifier cannot be modified to separate two classes with a nonlinear decision boundary.
- D. The logistic sigmoid function maps from a probability to a continuous value in $(-\infty, \infty)$.
- E. Don't know.

Solution 5.

- This is correct. Over-regularizing the weights ($\lambda \rightarrow +\infty$) pushes all the weights except the intercept to 0. So the optimization process minimizes the error by learning an intercept to predict always the majority class.
- This is wrong, since this formula can be only used in regularized linear regression with the sum of squared differences as the loss and the Euclidean norm as the regularizer for the weights.
- This is wrong, because we can consider non-linear transformations of the attributes that allows to get non-linear decision boundaries.

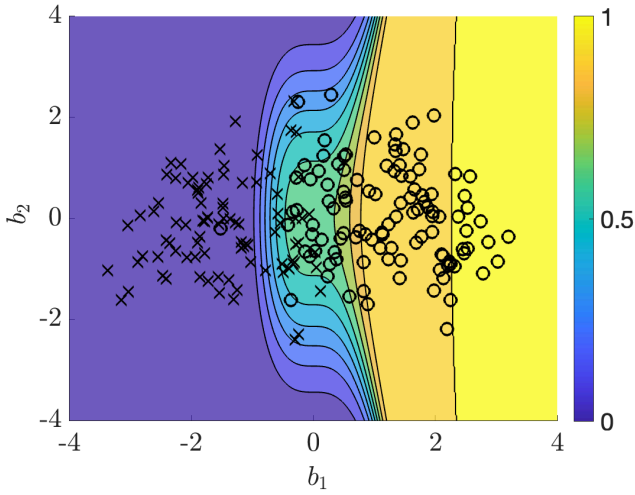


Figure 3: The logistic regression classifier, where the probability $p(y = \text{High} \mid \mathbf{b})$ is shown.

- This is wrong, as the logistic sigmoid maps the real line $\mathbb{R}$ to the interval $(0, 1)$.

Question 6. We consider the projection of the CCPP dataset on the first two principal components, and a logistic regression classifier where we use nonlinear transformations on the projection $\mathbf{b} = [b_1 \ b_2]^T$. The resulting classifier assigns for each $\mathbf{b}$ the probability $p(y = \text{High} \mid \mathbf{b}) = \sigma(\mathbf{w}^T \tilde{\mathbf{b}})$ as shown in Figure 3.

Which one of the following combination of transformation and weight vector corresponds to this classifier?

- A. $\mathbf{w} = [0.31 \ -0.06 \ 0.07]^T$
 $\tilde{\mathbf{b}} = [1 \ b_1^2 \ b_2^3]^T$
- B. $\mathbf{w} = [0.72 \ 3.13 \ -0.25]^T$
 $\tilde{\mathbf{b}} = [1 \ b_1^2 \ b_2^2]^T$
- C. $\mathbf{w} = [0.31 \ -0.06 \ 0.07]^T$
 $\tilde{\mathbf{b}} = [1 \ b_1^3 \ b_2^2]^T$
- D.** $\mathbf{w} = [0.72 \ 3.13 \ -0.25]^T$
 $\tilde{\mathbf{b}} = [1 \ b_1^3 \ b_2^2]^T$
- E. Don't know.

Solution 6.

One solution is to plot all the classifiers using a 2-dimensional grid and find the correct match (see Figure 4. The second option is to consider some test points $\mathbf{x} = [2, 0]$ and $\mathbf{x} = [-2 \ 0]$, and evaluate the classifiers. Note that the point $\mathbf{x} = [2 \ 0]$ is not enough as it give roughly the same result for two options.

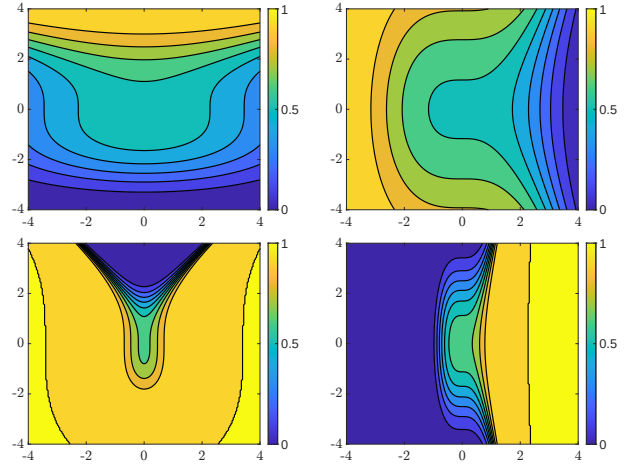


Figure 4: The result of each combination

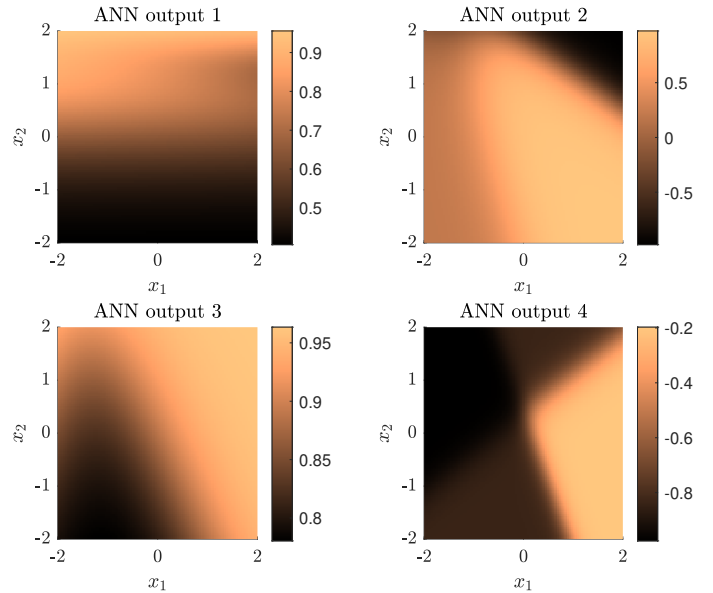


Figure 5: Suggested outputs of an ANN trained on the two attributes x_1 and x_2 from the CCPP dataset to predict the attribute x_3 .

Question 7. Consider an artificial neural network (ANN) trained on the CCPP dataset described in Table 1 to predict the continuous attribute x_3 from the two attributes x_1 and x_2 . Suppose the neural network takes the form:

$$f(x, \mathbf{w}) = h^{(2)} \left(w_0^{(2)} + \sum_{j=1}^2 w_j^{(2)} h^{(1)}([1 \ x_1 \ x_2] \mathbf{w}_j^{(1)}) \right).$$

where the activation functions are selected as $h^{(1)}(x) = \tanh(x)$ (the hyperbolic tangent activation function) and $h^{(2)}(x) = \tanh(x)$ (the hyperbolic tangent activation function) and the weights are given as:

$$\mathbf{w}_1^{(1)} = \begin{bmatrix} 2.2 \\ 0.7 \\ -0.3 \end{bmatrix}, \quad \mathbf{w}_2^{(1)} = \begin{bmatrix} -0.2 \\ 0.8 \\ 0.4 \end{bmatrix},$$

$$\mathbf{w}^{(2)} = \begin{bmatrix} -0.7 \\ 0.5 \end{bmatrix}, \quad w_0^{(2)} = 2.2.$$

Which one of the functions in Figure 5 will then correspond to the function f ?

- A. ANN output 1
- B. ANN output 2
- C. ANN output 3**
- D. ANN output 4
- E. Don't know.

Solution 7.

It suffices to compute the activation of the neural network at $[x_1 \ x_2] = [-2.0 \ -1.88]$. The activation of each of the two hidden neurons is:

$$n_1 = h^{(1)}([1.0 \ -2.0 \ -1.88] \mathbf{w}_1^{(1)}) = 0.877$$

$$n_2 = h^{(1)}([1.0 \ -2.0 \ -1.88] \mathbf{w}_2^{(1)}) = -0.988.$$

The final output is then computed as:

$$f(x, \mathbf{w}) = h^{(2)} \left(w_0^{(2)} + \sum_{j=1}^2 w_j^{(2)} h^{(1)}([1 \ x_1 \ x_2] \mathbf{w}_j^{(1)}) \right)$$

$$= h^{(2)} \left(w_0^{(2)} + \sum_{j=1}^2 w_j^{(2)} n_j \right) = 0.798.$$

This rules out all options except C.

	f_1	f_2	f_3	f_4
o_1	0	1	0	0
o_2	0	1	0	0
o_3	0	1	0	1
o_4	0	1	0	0
o_5	1	1	0	1
o_6	0	1	0	1
o_7	0	0	0	1
o_8	0	0	0	1
o_9	0	0	0	1
o_{10}	0	0	0	1

Table 2: Binarized version of the CCPP dataset. Each of the features f_i are obtained by taking an attribute x_i and letting $f_i = 1$ correspond to a value x_i greater than the median (otherwise $f_i = 0$). The colors indicate classes such that the black observations $\{o_1, o_2, o_3, o_4, o_5, o_6\}$ belong to class Low and the red observations $\{o_7, o_8, o_9, o_{10}\}$ to class High.

Question 8. Consider a new, binarized observation

$$o_{11} = [0 \ ? \ ? \ 1]^T,$$

which contains two unknown values each indicated with a ?. What is the minimum and maximum achievable value of the Simple Matching Coefficient (SMC) between o_6 from Table 2 and the new observation o_{11} ?

- A.** $\min \text{SMC}(o_6, o_{11}) = 0.50$
 $\max \text{SMC}(o_6, o_{11}) = 1.0$
- B. $\min \text{SMC}(o_6, o_{11}) = 0.75$
 $\max \text{SMC}(o_6, o_{11}) = 0.9$
- C. $\min \text{SMC}(o_6, o_{11}) = 0.25$
 $\max \text{SMC}(o_6, o_{11}) = 1.0$
- D. $\min \text{SMC}(o_6, o_{11}) = 0.25$
 $\max \text{SMC}(o_6, o_{11}) = 0.9$
- E. Don't know.

Solution 8. The Simple Matching Coefficient is computed as $\text{SMC} = (f_{11} + f_{00})/M$

Max: Trivially the max would be 1 for $o_{11} = o_6$, i.e. $o_{11} : [0 \ 1 \ 0 \ 1]$, as SMC only depends on the elements where the two vectors agree.

Min: The min is obtained by minimising f_{11} and f_{00} , i.e. $o_{11} = \neg o_6$ where possible, i.e.

$$o_{11} : [0 \ 0 \ 1 \ 1]$$

Outer fold $i=1$						
Inner folds	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$	Test
Model 1	0.12	0.21	0.22	0.23	0.15	0.24
Model 2	0.30	0.11	0.15	0.30	0.28	0.17
Model 3	0.21	0.14	0.26	0.17	0.26	0.22

Outer fold $i=2$						
Inner folds	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$	Test
Model 1	0.28	0.18	0.19	0.27	0.12	0.19
Model 2	0.16	0.20	0.27	0.30	0.25	0.16
Model 3	0.13	0.16	0.21	0.17	0.13	0.25

Table 3: The information for the two-level cross-validation. For each outer fold we are given the validation error for each model on the inner folds, together with the test error on the associated partition of the outer fold.

giving an SMC of 2/4.

Question 9. We consider a subset of $N = 1000$ points from the CCPP dataset and a classifier with three different complexity levels (Model 1, Model 2, Model 3). We apply two-level cross-validation to estimate the generalization error, with 2 outer folds and 5 inner folds. In Table 3 the information for each outer fold, model, and inner fold is given, together with the performance of each model on the test partition of the outer fold.

Which one the following options is the correct estimate of the generalization error?

- A. $\hat{E}^{\text{gen}} = 0.165$
- B. $\hat{E}^{\text{gen}} = 0.245$
- C. $\hat{E}^{\text{gen}} = 0.215$
- D. $\hat{E}^{\text{gen}} = 0.205$
- E. Don't know.

Solution 9. For each outer fold, we compute the average for each Model using the information provided for the inner folds, and we then select the Model with the smallest result. From the outer fold 1, we select the Model 1 with 0.24 Test. From the outer fold 2, we select the Model 3 with 0.25 Test. So the estimated test error is $0.5 \cdot (0.24 + 0.25) = 0.245$.

Question 10. Which one of the following statements

$p(x_1, x_3 y)$	$y = \text{Low}$	$y = \text{High}$
$x_1 = 0, x_3 = 0$	0.04	0.25
$x_1 = 0, x_3 = 1$	0.03	0.68
$x_1 = 1, x_3 = 0$	0.66	0.02
$x_1 = 1, x_3 = 1$	0.27	0.05

Table 4: The class conditional probabilities for the binarized attributes x_1, x_3 of the CCPP dataset.

regarding artificial neural networks is *correct*?

- A. **An artificial neural network with linear activation functions ($h^{(i)}(x) = x$) can be written in terms of a linear regression model.**
- B. We are guaranteed to find the optimal artificial neural network every time we train the model from a random initialization.
- C. The optimal number of hidden units of an artificial neural network should be selected in terms of the model that minimizes the training error.
- D. Regularizing an artificial neural network regression model by penalizing the weights using the squared Euclidean norm and a positive value λ decreases bias and increases variance.
- E. Don't know.

Solution 10.

- Correct. Essentially, if we have no activation functions, then the layers constitute only matrix multiplications, so the final model is equivalent to a linear regression model.
- This is wrong. The ANN have many local/global minima, and it is a highly-non-convex optimization problem. So we cannot guarantee that we converge to an optimal model every time we train a model starting from a random initialization.
- This is wrong. We can estimate the optimal number of units using cross-validation.
- This is wrong. In general, including a regularizer that penalizes the weights aims to reduce the variance while introducing some bias.

Question 11. Consider the Table 4 that shows the class conditional joint probability for the attributes

x_1, x_3 of the CCPP dataset after binarization, while the prior probabilities for the two classes are $p(y = \text{Low}) = 0.53$ and $p(y = \text{High}) = 0.47$.

What is the probability the energy production to be High when $x_1 = 0$?

- A. $p(y = \text{High} \mid x_1 = 0) \approx 0.92$
- B. $p(y = \text{High} \mid x_1 = 0) \approx 0.82$
- C. $p(y = \text{High} \mid x_1 = 0) \approx 0.56$
- D. $p(y = \text{High} \mid x_1 = 0) \approx 0.78$
- E. Don't know.

Solution 11. We have to compute the probability $p(y = 1 \mid x_1 = 0) = p(x_1 = 0 \mid y = 1)p(y = 1)/p(x_1 = 0)$. We compute first the class conditionals

$$p(x_1 = 0 \mid y = 1) = p(x_1 = 0, x_2 = 0 \mid y = 1) + p(x_1 = 0, x_2 = 1 \mid y = 1) = 0.93$$

and

$$p(x_1 = 0 \mid y = 0) = p(x_1 = 0, x_2 = 0 \mid y = 0) + p(x_1 = 0, x_2 = 1 \mid y = 0) = 0.07$$

and then we compute the marginal

$$\begin{aligned} p(x_1 = 0) &= p(x_1 = 0 \mid y = 1)p(y = 1) \\ &\quad + p(x_1 = 0 \mid y = 0)p(y = 0) \\ &= 0.93 \cdot 0.47 + 0.07 \cdot 0.53 = 0.4742 \end{aligned}$$

so finally we can compute

$$\begin{aligned} p(y = 1 \mid x_1 = 0) &= \frac{p(x_1 = 0 \mid y = 1)p(y = 1)}{p(x_1 = 0)} \\ &= \frac{0.93 \cdot 0.47}{0.4742} \approx 0.92 \end{aligned}$$

	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$
y_i	-2.9	-0.4	0.7	2.5	4.5
x_i	-3.4	-1.3	0.5	2.4	4.2

Table 5: The synthetic dataset for the regularized linear regression model.

Question 12. We consider the synthetic dataset given in Table 5 where x represents the one dimensional input and y the corresponding one dimensional output. We

standardize the input before learning the optimal bias w_0^* and parameter w_1^* for a regularized linear regression model with $\lambda = 0.7$, where we consider the formulation from Section 14.1.

What is the prediction of the trained model for the input x_2 ?

- A. $\hat{y} \approx -0.76$
- B. $\hat{y} \approx -0.54$
- C. $\hat{y} \approx -0.65$
- D. $\hat{y} \approx -2.22$
- E. Don't know.

Solution 12. First we need to standardize the input with $\mu = 0.48$ and $\sigma \approx 2.9895$ to get $\tilde{x}_i = (x_i - \mu)/\sigma$. Then we apply the formula for finding the optimal parameters either by $w_0^* = \frac{1}{5} \sum_i y_i$ and $w_1^* = \sum_{i=1} \tilde{x}_i (y_i - w_0^*) / (\sum_{i=1} \tilde{x}_i^2 + \lambda)$ or by applying the general formula (from Sec. 14.1) with $\tilde{\mathbf{X}} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ \tilde{x}_1 & \tilde{x}_2 & \tilde{x}_3 & \tilde{x}_4 & \tilde{x}_5 \end{bmatrix}^T$ as follows $\mathbf{w}^* = (\tilde{\mathbf{X}}^T \tilde{\mathbf{X}} + \lambda \cdot \text{diag}([0, 1]))^{-1} \tilde{\mathbf{X}}^T \mathbf{y}$, note that the intercept should not be regularized. The result is in both cases $w_0^* = 0.88$ and $w_1^* = 2.3855$. We can then predict with our model as $\hat{y} = w_1^* \tilde{x} + w_0^* = w_1^*(x - \mu)/\sigma + w_0^*$. Note that the input to the trained model must be standardized as the parameters correspond to the standardized training input.

Question 13. We consider the class conditional distributions for the x_1, x_3 attributes of the CCPP dataset which we model with the following multivariate normal distributions

$$p(x_1, x_3|y=1) = \mathcal{N}(\boldsymbol{\mu}^{(1)}, \boldsymbol{\Sigma}^{(1)}), \quad \text{where}$$

$$\boldsymbol{\mu}^{(1)} = \begin{bmatrix} 0.77 \\ -0.41 \end{bmatrix}, \quad \boldsymbol{\Sigma}^{(1)} = \begin{bmatrix} 0.29 & -0.12 \\ -0.12 & 0.55 \end{bmatrix},$$

$$p(x_1, x_3|y=2) = \mathcal{N}(\boldsymbol{\mu}^{(2)}, \boldsymbol{\Sigma}^{(2)}), \quad \text{where}$$

$$\boldsymbol{\mu}^{(2)} = \begin{bmatrix} -0.91 \\ 0.5 \end{bmatrix}, \quad \boldsymbol{\Sigma}^{(2)} = \begin{bmatrix} 0.32 & -0.11 \\ -0.11 & 1.12 \end{bmatrix}.$$

The prior probabilities for the classes Low and High are $p(y=1) = 0.53$ and $p(y=2) = 0.47$ respectively. We are given a new test point $\mathbf{x}_{\text{test}} = [0 \ 0.7]^T$ to classify and we consider a *Naive Bayes* approach.

Which one of the following statements is *correct*?

- A. We do not have enough information to classify the point to one of the two classes.
- B. We will classify the point to the Low class ($y=1$).
- C. We will classify the point to the High class ($y=2$).**
- D. The point belongs in both classes with the same probability $p(y=1|\mathbf{x}_{\text{test}}) = p(y=2|\mathbf{x}_{\text{test}}) = 0.5$.
- E. Don't know.

Solution 13. We need to compute the ratio

$$\begin{aligned} \frac{p(y=1|\mathbf{x}_{\text{test}})}{p(y=2|\mathbf{x}_{\text{test}})} &= \frac{p(\mathbf{x}_{\text{test}}|y=1)p(y=1)/p(\mathbf{x}_{\text{test}})}{p(\mathbf{x}_{\text{test}}|y=2)p(y=2)/p(\mathbf{x}_{\text{test}})} \\ &= \frac{\mathcal{N}(\mathbf{x}_{\text{test}}|\boldsymbol{\mu}^{(1)}, \text{diag}[\boldsymbol{\Sigma}^{(1)}])p(y=1)}{\mathcal{N}(\mathbf{x}_{\text{test}}|\boldsymbol{\mu}^{(2)}, \text{diag}[\boldsymbol{\Sigma}^{(2)}])p(y=2)} \\ &\approx 0.73 < 1, \end{aligned}$$

so the point will be classified to class “High” ($y=2$). Note that we considered only the diagonal elements of the covariances $\text{diag}[\boldsymbol{\Sigma}^{(1)}]$ due to the Naive Bayes assumption, which corresponds to the product of the two normal distributions $\mathcal{N}(x_1|\mu_1^{(1)}, (\sigma_1^2)^{(1)}) \cdot \mathcal{N}(x_3|\mu_3^{(1)}, (\sigma_3^2)^{(1)})$ and similarly for $y=2$.

Question 14. Which one of the following statements regarding ensemble methods is *correct*?

- A. If the accuracy of a classifier in AdaBoost is lower than 0.5, then this classifier will have no influence on the result.
- B. In AdaBoost with a regularized logistic regression classifier as the weak learner, increasing the number of iterations increases the chance of overfitting.**
- C. In bagging and boosting observations are sampled without replacement, but in boosting misclassified observations are given a higher probability of being sampled in the next round.
- D. In AdaBoost the more errors a classifier makes in a given round of boosting the more the weights will be increased for the wrongly classified observations.
- E. Don't know.

Solution 14.

- This is wrong. Only when error is 0.5 the classifier has no influence.
- Correct. In AdaBoost increasing the number of iterations of the algorithm increases the chances of overfitting even if the weak learning is simply a linear classifier.
- This is wrong. For bagging and boosting we sample *with* replacement.
- This is wrong. The less errors a classifier makes the more the weights of the misclassified points will be increased.

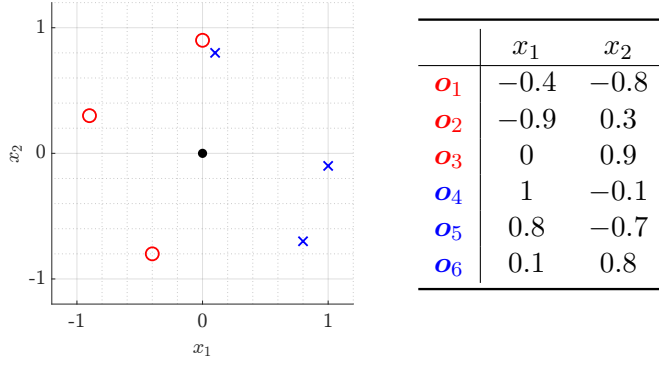


Figure 6: The synthetic dataset, where the colors indicate the classes such that the red observations $\{\mathbf{o}_1, \mathbf{o}_2, \mathbf{o}_3\}$ belong to class C_1 , and the blue observations $\{\mathbf{o}_4, \mathbf{o}_5, \mathbf{o}_6\}$ belong to class C_2 .

Question 15. We are given the synthetic dataset in Figure 6 where the color indicates the class. We consider the p -distances d_2 , d_1 , d_∞ and the KNN classifier with $K = 3$ to predict the label of the test point $\mathbf{o}_{\text{test}} = [0 \ 0]^T$.

Which one of the following statements is *correct*?

- A. d_2 and d_∞ predict the same label.
- B. d_2 and d_1 predict the same label.
- C. d_1 and d_∞ predict the same label.**
- D. All the norms predict the same result
- E. Don't know.

Solution 15. We need to compute the $d_p(\mathbf{x}_{\text{test}}, \mathbf{x}_i)$, $i = 1, \dots, 6$ for each $p = 1, 2, \infty$. We can then find the 3-NN for each scenario, and see that the d_1 and d_∞ predict the same label.

Question 16. We consider three different classifiers (Model 1, Model 2, Model 3) and the McNemar test is used to compare M1 vs. M2 and M1 vs. M3. In Figure 7 we are given four options of Beta distributions together with the associated information for the comparisons i.e., such that M1 implies that Model 1 predicts correctly and $\overline{\text{M1}}$ that it predicts incorrectly.

For which one of the four options does the Beta

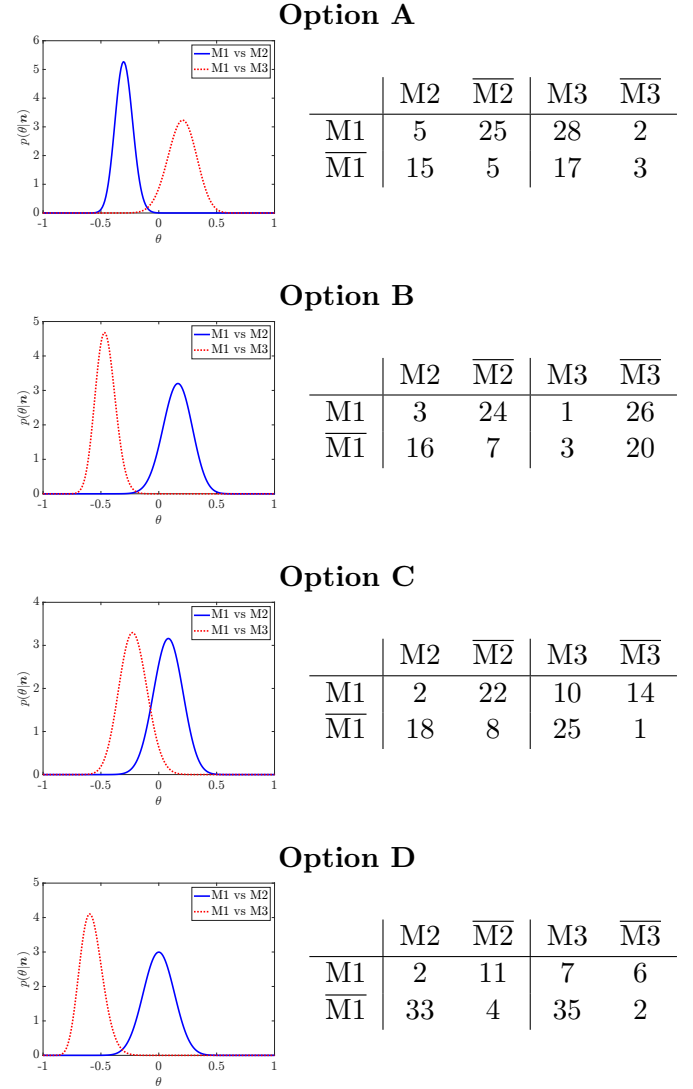


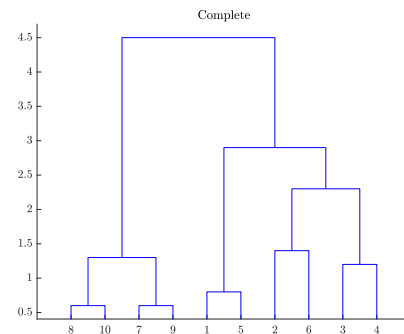
Figure 7: Four options of McNemar tests comparing M1 vs. M2 and M1 vs. M3. For each option we are given two Beta distributions together with a table that provides the information for the comparisons. Note that M1 implies that Model 1 predicts correctly and $\overline{\text{M1}}$ that it predicts incorrectly.

distributions match the information in the table?

- A. Option A
- B. Option B
- C. Option C**
- D. Option D
- E. Don't know.

Solution 16. Only Option C gives the correct probability density functions for the provided information. For Option A, blue corresponds to M1 vs. M3 and red to M1 vs. M2. For Option B, the correct red for M1 vs. M3 should have been centered around 0. as the M1 is better than M3. For Option D, blue implies that M1 performs similar to M2 which does not agree with the provided info.

Figure 8: SOLUTION: Complete linkage



Question 17. Consider the observations and the pairwise distances in Table 6. At which height will groups/clusters containing o_9 and o_{10} merge in a dendrogram when using the *complete* linkage function?

- A. 1.3**
- B. 0.6
- C. 0.9
- D. 1.4
- E. Don't know.

Solution 17. Groups containing o_9 and o_{10} will merge at 1.3 as seen from the dendrogram shown in Figure 8.

Question 18. Consider again the CCPP dataset of Table 1. Suppose we limit ourselves to $N = 6$ observations from the original dataset and furthermore suppose we consider only the features x_1 and x_4 to predict the class $y = 1$ or $y = 2$. We use a KNN

	o_1	o_2	o_3	o_4	o_5	o_6	o_7	o_8	o_9	o_{10}
o_1	0.0	2.6	2.8	2.1	0.8	1.7	3.9	3.8	4.1	4.2
o_2	2.6	0.0	1.8	2.3	2.7	1.4	1.9	1.7	2.0	2.2
o_3	2.8	1.8	0.0	1.2	2.9	2.1	2.2	1.6	2.3	2.0
o_4	2.1	2.3	1.2	0.0	2.4	1.8	2.6	2.5	3.0	2.9
o_5	0.8	2.7	2.9	2.4	0.0	2.0	4.2	4.0	4.4	4.5
o_6	1.7	1.4	2.1	1.8	2.0	0.0	2.4	2.6	2.7	3.0
o_7	3.9	1.9	2.2	2.6	4.2	2.4	0.0	1.2	0.6	1.3
o_8	3.8	1.7	1.6	2.5	4.0	2.6	1.2	0.0	1.0	0.6
o_9	4.1	2.0	2.3	3.0	4.4	2.7	0.6	1.0	0.0	0.9
o_{10}	4.2	2.2	2.0	2.9	4.5	3.0	1.3	0.6	0.9	0.0

Table 6: The pairwise Euclidian distances, $d(o_i, o_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2 = \sqrt{\sum_{k=1}^M (x_{ik} - x_{jk})^2}$ between 10 observations from the CCPP dataset (recall that $M = 4$). Each observation o_i corresponds to a row of the data matrix $\mathbf{X}$ of Table 1. The colors indicate classes such that the black observations $\{o_1, o_2, o_3, o_4, o_5, o_6\}$ belong to class C_1 (corresponding to Low output), and the red observations $\{o_7, o_8, o_9, o_{10}\}$ belong to class C_2 (corresponding to High output). To avoid single features to dominate, the dataset was standardized by subtracting the mean and dividing by the standard deviation.

Variable	y^{true}	$t = 1$
y_1	2	2
y_2	2	2
y_3	1	2
y_4	2	1
y_5	1	2
y_6	1	2

Table 7: For each of the $N = 6$ observations (first column), the table indicate the true class labels y^{true} (second column) and the predicted outputs of the AdaBoost classifier (third column) for the first round of boosting $t = 1$.

classification model ($K = 1$) to this dataset and apply AdaBoost to improve the performance. After the first round of boosting, we obtain predictions and the true class labels as tabulated in Table 7. Note that $\log(x)$ consistently refers to the natural logarithm (i.e., base e). Given this information, how will the AdaBoost update the weights $\mathbf{w}$?

A. $[0.112 \quad 0.112 \quad 0.194 \quad 0.194 \quad 0.194 \quad 0.194]$

B. $[0.25 \quad 0.25 \quad 0.125 \quad 0.125 \quad 0.125 \quad 0.125]$

C. $[0.059 \quad 0.059 \quad 0.221 \quad 0.221 \quad 0.221 \quad 0.221]$

D. $[0.341 \quad 0.341 \quad 0.079 \quad 0.079 \quad 0.079 \quad 0.079]$

E. Don't know.

Solution 18.

We first observe the AdaBoost classifier at $t = 1$ mis-classify observations:

$$\{y_3, y_4, y_5, y_6\}$$

Since the weights are just $w_i = \frac{1}{N}$, we therefore get:

$$\epsilon_{t=1} = \sum_i w_i(t)(1 - \delta_{f_t(x_i), y_i}) = 0.667$$

From this, we compute α_t as

$$\alpha_t = \frac{1}{2} \log \frac{1 - \epsilon_t}{\epsilon_t} = -0.347$$

Scaling the observations corresponding to the mis-classified weights as $w_i e^{\alpha_t}$ and those corresponding to the correctly classified weights as $w_i e^{-\alpha_t}$ and normalizing the new weights to sum to one then give answer B.

Question 19. Consider the distances in Table 6 based on 10 observations from the CCPP dataset. The class labels C_1, C_2 (see table caption for details) will be predicted using a k -nearest neighbour classifier based on the distances given in Table 6 (ties are broken in the usual manner by considering the nearest observation from the tied classes). Suppose we use leave-one-out cross validation (i.e., the observation that is being predicted is left out) and a 3-nearest neighbour classifier

(i.e., $k = 3$). What is the accuracy computed for all $N = 10$ observations?

- A. accuracy = $\frac{1}{5}$
- B. accuracy = $\frac{3}{5}$
- C. accuracy = $\frac{4}{5}$
- D. accuracy = 1**
- E. Don't know.

Solution 19.

The correct answer is D. To see this, recall that leave-one-out cross-validation means we train a total of $N = 10$ models, each model being tested on a single observation and trained on the remaining such that each observation is used for testing exactly once.

The model considered is KNN classifier with $k = 3$. To figure out the accuracy for a particular observation i (i.e., the test set for this fold), we train a model on the other observations and predict on observation i . To do that, simply find the observation different than i closest to i according to Table 6 and predict i as belonging to it's class. Concretely, we find: $N(o_1, k) = \{o_5, o_6, o_4\}$, $N(o_2, k) = \{o_6, o_8, o_3\}$, $N(o_3, k) = \{o_4, o_8, o_2\}$, $N(o_4, k) = \{o_3, o_6, o_1\}$, $N(o_5, k) = \{o_1, o_6, o_4\}$, $N(o_6, k) = \{o_2, o_1, o_4\}$, $N(o_7, k) = \{o_9, o_8, o_{10}\}$, $N(o_8, k) = \{o_{10}, o_9, o_7\}$, $N(o_9, k) = \{o_7, o_{10}, o_8\}$, and $N(o_{10}, k) = \{o_8, o_9, o_7\}$.

The accuracy is then found by observing how often the class label of the observation in the neighborhood agrees with the true class label. We find this happens for observations

$$\{o_1, o_2, o_3, o_4, o_5, o_6, o_7, o_8, o_9, o_{10}\}$$

and the remaining observations are therefore erroneously classified, in other words, the accuracy is $\frac{10}{10}$.

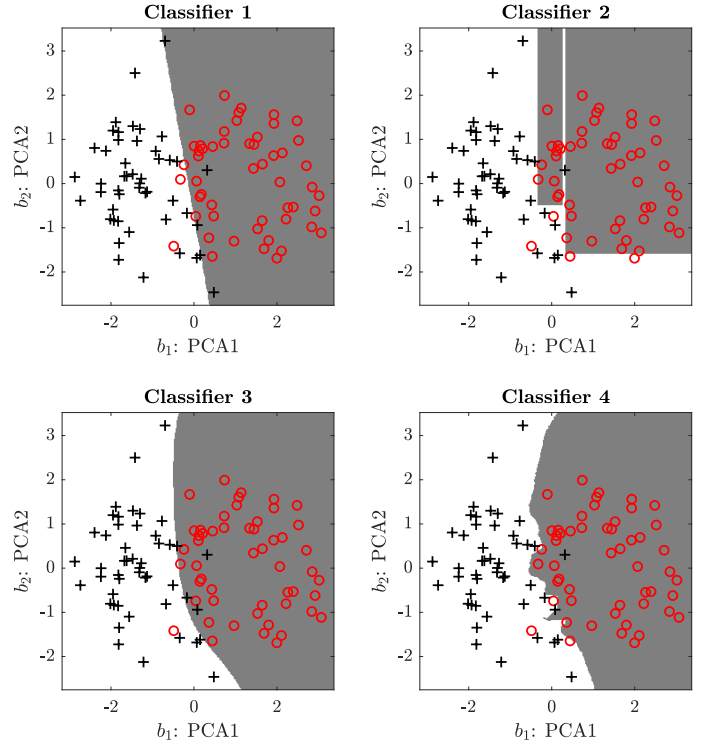


Figure 9: Decision boundaries for four different classifiers trained on the CCP dataset when projected onto the first two principal components.

Question 20. We will consider a subset of the CCP dataset (described in Table 1) after it has been projected onto the first two principal components b_1 and b_2 , thereby giving rise to a smaller two-dimensional dataset. We will consider the following four classifiers:

LREG: Logistic regression

ANN: Artificial neural network with 5 hidden units

CT: Classification tree with regular axis-aligned splits ($b_i < c$)

KNN: K-nearest neighbours with $K = 3$

Suppose the classifiers are trained on the two-dimensional dataset and the decision boundary for each of the four classifiers is given in Figure 9. Which one of the following statements is correct?

- A. Classifier 1 corresponds to **KNN**,
Classifier 2 corresponds to **ANN**,
Classifier 3 corresponds to **LREG**,
Classifier 4 corresponds to **CT**.
- B. Classifier 1 corresponds to **LREG**,

Classifier 2 corresponds to **ANN**,
 Classifier 3 corresponds to **KNN**,
 Classifier 4 corresponds to **CT**.

- C. Classifier 1 corresponds to **CT**,
 Classifier 2 corresponds to **LREG**,
 Classifier 3 corresponds to **KNN**,
 Classifier 4 corresponds to **ANN**.

**D. Classifier 1 corresponds to LREG,
 Classifier 2 corresponds to CT,
 Classifier 3 corresponds to ANN,
 Classifier 4 corresponds to KNN.**

- E. Don't know.

Solution 20. To solve this problem, we have to use our intuition about what the typical decision boundaries for the different methods look like:

- A KNN method will have decision boundaries dictated by the nearest neighbors. That is, points (x, y) where the nearest K neighbors are in one class must be in the same class and therefore the boundaries will be fairly complex and respect the data distribution well.
- A decision tree has axis aligned splits, therefore the boundaries must be vertical or horizontal
- A logistic regression model must have linear boundaries
- An artificial neural network with few hidden units can have some non-linearity, but otherwise have boundaries of limited complexity and consisting of relatively simple shapes

It is easy to see this rules out all but option D.

Question 21. Consider the synthetic 1D dataset with four training observations:

$$\mathbf{X} = \{-3, -1, 5, 6\}.$$

A standard kernel density estimator (KDE) with $\sigma^2 = 0.5$ is used to determine if test observations are anomalous based on their estimated densities. An observation will be flagged as an anomaly if $p(x) < 0.015$. We test two new observations located at $x = -4$ and $x = 2$.

Determine which one of the following statements is correct.

- A. Both test observations are anomalies.
B. The observation at $x = -4$ is *not* an anomaly but the observation at $x = 2$ is an anomaly.
 C. The obseravtion at $x = -4$ is an anomaly but the observation at $x = 2$ is *not* an anomaly.
 D. Neither of the test observations are anomalies.
 E. Don't know.

Solution 21.

The correct answer can be identified from the solution shown in ?? 21, i.e.,

The full KDE is not required and the density at the test points can be found from:

$$\begin{aligned} p(x = -4) &= \frac{1}{4 \cdot \sqrt{\pi}} \times \\ &\quad \left(\exp\left(-(-4 - (-3))^2\right) + \right. \\ &\quad \exp\left(-(-4 - (-1))^2\right) + \\ &\quad \exp\left(-(-4 - 5)^2\right) + \\ &\quad \left. \exp\left(-(-4 - 6)^2\right) \right) \\ &\approx .0293 > 0.015 \end{aligned}$$

$$\begin{aligned} p(x = 2) &= \frac{1}{4 \cdot \sqrt{\pi}} \times \\ &\quad \left(\exp\left(-(2 - (-3))^2\right) + \right. \\ &\quad \exp\left(-(2 - (-1))^2\right) + \\ &\quad \exp\left(-(2 - 5)^2\right) + \\ &\quad \left. \exp\left(-(2 - 6)^2\right) \right) \\ &\approx 0 < 0.015 \end{aligned}$$

Note: confusing σ with σ^2 will result in both test observations being anomalies.

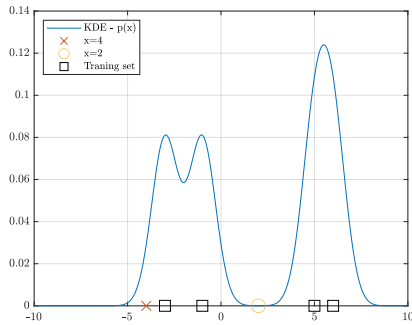


Figure 10: SOLUTION: KDE

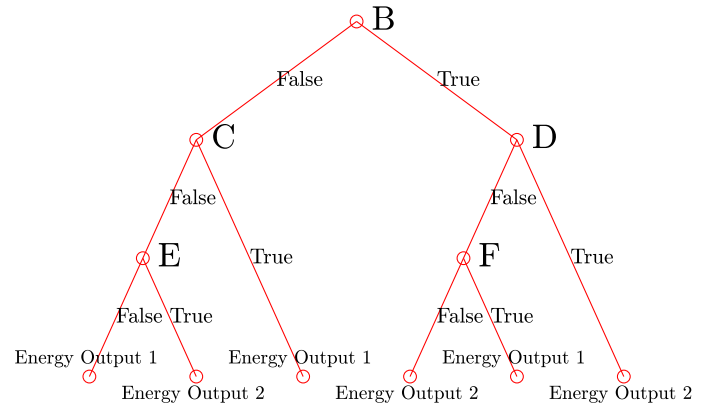


Figure 11: Structure of decision tree. The goal is to determine the splitting rules.

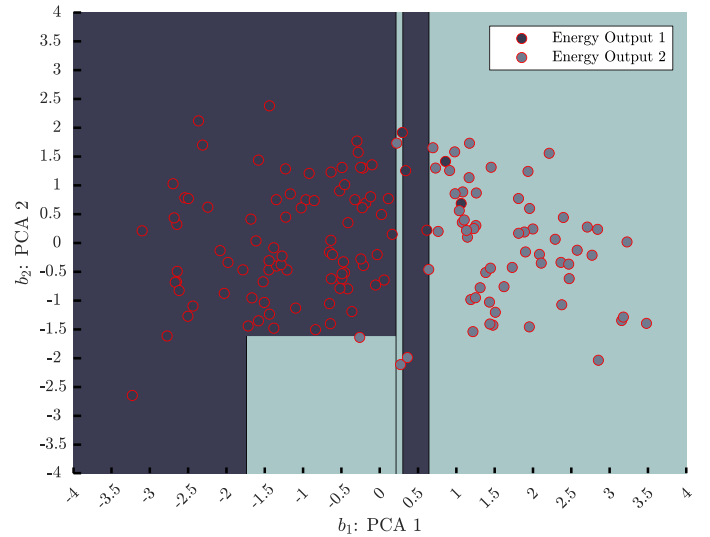


Figure 12: Classification boundary.

Question 22. We will consider the CCPP dataset projected onto the first two principal directions. Suppose we train a decision tree to predict which of the two classes an observation belongs to. Since the attributes are continuous, we will consider binary splits of the form $b_i \geq z$ for different values of i and z , where b_1, b_2 refer to the coordinates of the observations when projected onto principal directions. Suppose the trained decision tree has the form shown in Figure 11, and that according to the tree the predicted label assignment for a subset of the observations is as given in Figure 12, what is then the correct rule assignment to the nodes

in the decision tree?

- A. **B:** $b_1 \geq 0.63$, **C:** $b_1 \geq 0.28$, **D:** $b_2 \geq -1.63$, **E:** $b_1 \geq 0.19$, **F:** $b_1 \geq -1.75$
- B. **B:** $b_1 \geq -1.75$, **C:** $b_2 \geq -1.63$, **D:** $b_1 \geq 0.28$, **E:** $b_1 \geq 0.63$, **F:** $b_1 \geq 0.19$
- C.** **B:** $b_1 \geq 0.19$, **C:** $b_2 \geq -1.63$, **D:** $b_1 \geq 0.63$, **E:** $b_1 \geq -1.75$, **F:** $b_1 \geq 0.28$
- D. **B:** $b_2 \geq -1.63$, **C:** $b_1 \geq 0.63$, **D:** $b_1 \geq -1.75$, **E:** $b_1 \geq 0.28$, **F:** $b_1 \geq 0.19$
- E. Don't know.

Solution 22.

This problem is solved by using the definition of a decision tree and observing what classification rule each of the assignment of features to node names in the decision tree will result in. I.e. beginning at the top of the tree, check if the condition assigned to the node is met and proceed along the true or false leg of the tree.

The resulting decision boundaries for each of the options are shown in Figure 13 and it follows answer C is correct.

Question 23. Consider a two-dimensional data set comprised of $N = 9$ observations shown in Figure 14. The dataset consists of three classes indicated by the blue squares (class 1), red triangles (class 2) and yellow circles (class 3). In the figure, the decision boundaries for four K -nearest neighbor classifiers (KNN) are shown. Which one of the plots correspond to the $K = 3$ nearest-neighbour classifier assuming ties are broken by assigning to the *nearest* neighbour's class?

- A. KNN decision boundary A
- B. KNN decision boundary B
- C. KNN decision boundary C
- D. KNN decision boundary D**
- E. Don't know.

Solution 23. A point to the left of the figure must clearly be assigned to class 2 because of the two triangles. A point furthest to the right will have three different classes in its $K = 3$ nearest neighborhood and the tie is broken by assigning it to class 2 as well. The rules out all options except A and B. These differ only

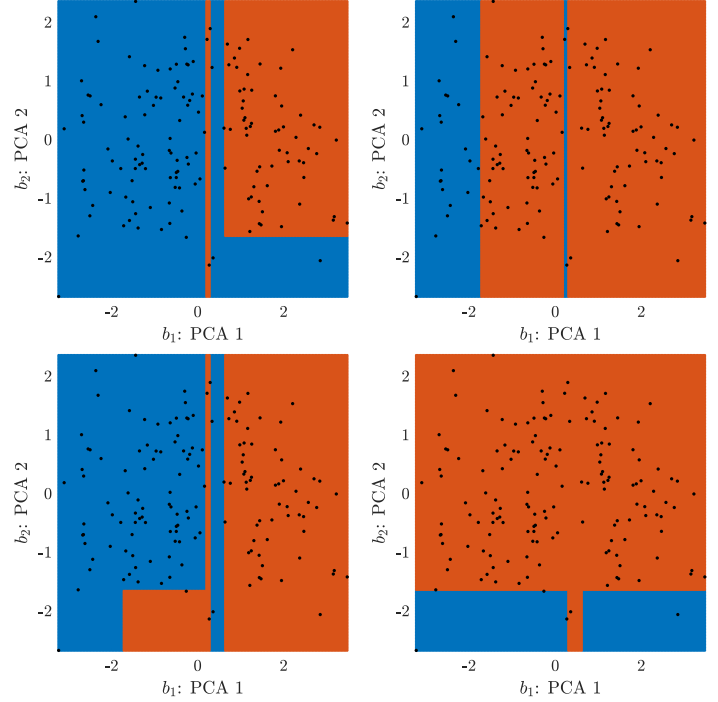


Figure 13: Classification trees induced by each of the options. (Top row: option A and B, bottom row: C and D)

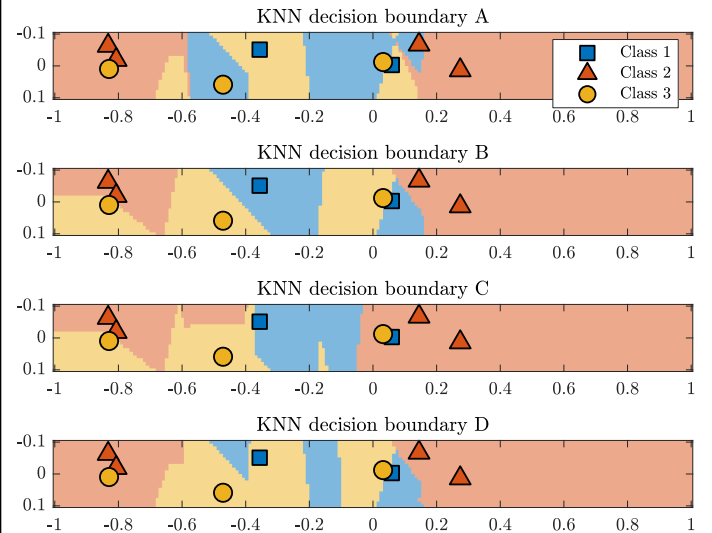


Figure 14: Decision boundaries for four KNN classifiers.

in the assignment of a point near the left-most blue square, but a point here will have two yellow circles in its $K = 3$ neighborhood and therefore belong to class 3. This rules out all options except D.

Question 24. A neural network classifier is trained to distinguish between two classes $y \in \{1, 2\}$ and produce class-probability $\hat{y}$ and the *receiver operator characteristic* (ROC) curve of the network when evaluated on a test set with $N = 8$ observations is shown in Figure 16. Suppose we plot the predictions on the $N = 8$ test observations by their $\hat{y}$ value along the x -axis and indicate the class labels by either a black circle (class $y = 1$) or red cross ($y = 2$), which one of the subplots in Figure 16 then corresponds to the ROC curve in Figure 15?

A. Prediction A

B. Prediction B

C. Prediction C

D. Prediction D

E. Don't know.

Solution 24. The correct answer is D. To see this, recall that the ROC curve is computed from the false positive rate (FPR) and true positive rate (TPR) for particular choices of threshold value $\hat{y}$. To compute e.g. the TPR, one assumes every observation predicted to belong to class 1 with a probability higher than $\hat{y}$ is actually assigned to class one. We then divide the total number of observations belonging to class one *and which are predicted to belong to class 1* with the number of observations in the *positive* class.

Similarly for the FPR, where we now count the number of observations that are assigned to class one *but in fact belongs to class 0*, divided by the total number of observations in the *negative* class.

This procedure is then repeated for different threshold values to obtain the curves shown in Figure 17. The ROC curve is then obtained by plotting these two curves against each other. I.e. for each threshold value, the point

$$(x, y) = (\text{FPR}, \text{TPR})$$

is on the AUC curve. This rules out all options except D. For completeness, we have included the ROC curves for all options in Figure 18.

Question 25. In the context of the Apriori algorithm for association rule mining, which one of the following

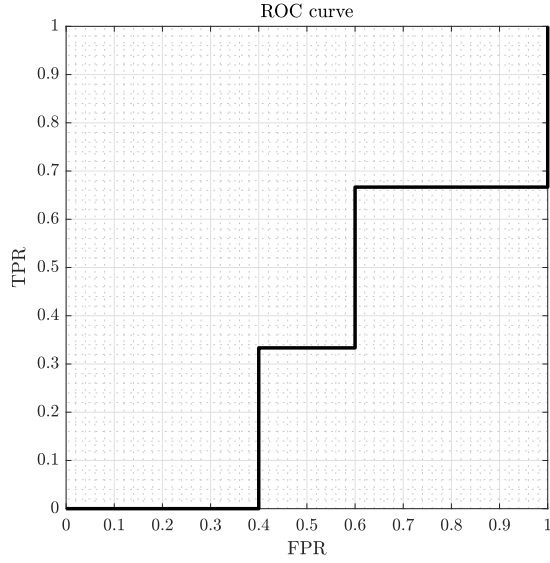


Figure 15: ROC curve for a neural network classifier, where the predictions and true class labels are one of the options in Figure 16.

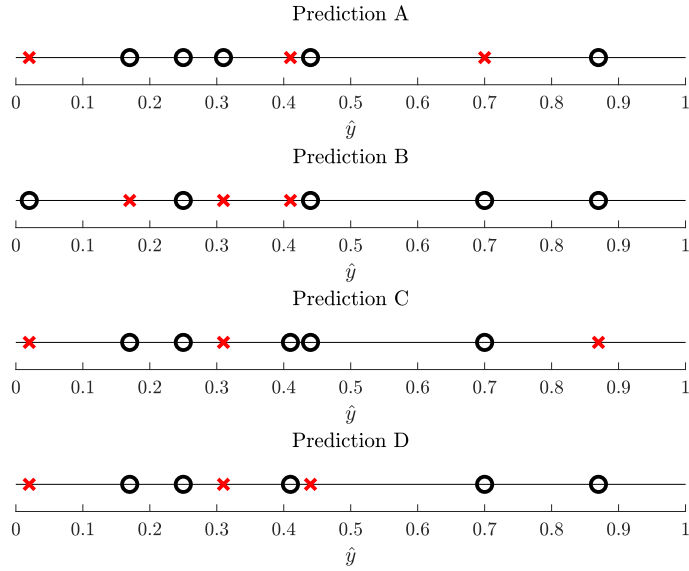


Figure 16: Four candidate predictions for the ROC curve in Figure 15. The observations are plotted horizontally, such that the position on the x -axis indicate the predicted probability $\hat{y} = p(y = 2|x)$, and the marker/color indicate the class membership, such that the black circles indicate the observation belongs to class $y_i = 1$ and red crosses to $y_i = 2$.

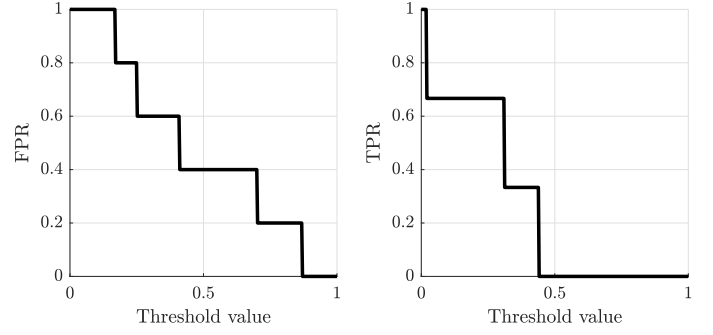


Figure 17: TPR, FPR curves for the classifier.

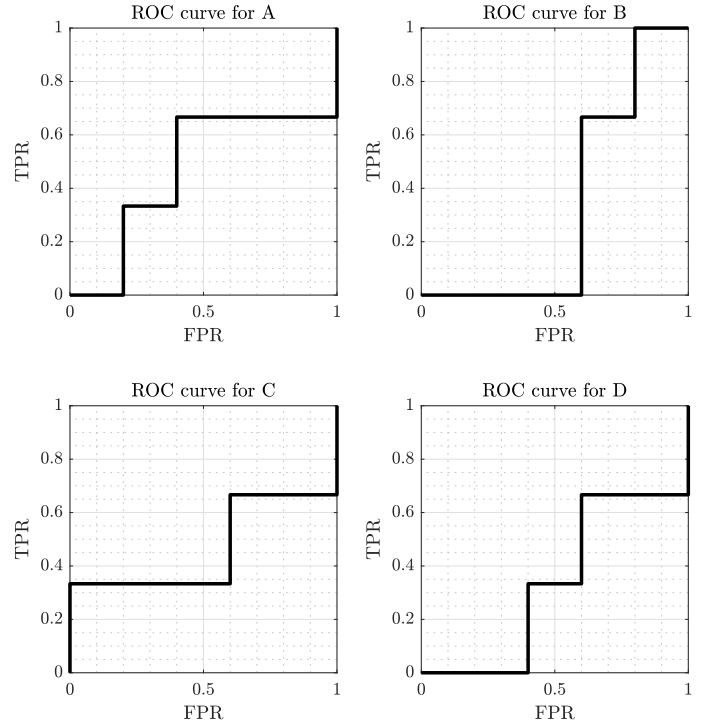


Figure 18: ROC curves for all options.

statements accurately describes a fundamental principle or step within the algorithm?

- A. Apriori randomly samples a candidate itemset and includes it in the list of frequent itemsets if its support is $\geq \epsilon$.
- B. Apriori prunes infrequent itemsets by exploiting the downward closure property of support, where any subset of a frequent itemset must also be frequent.**
- C. Apriori finds association rules with support $\geq \epsilon$ by exploiting that itemsets with $N + 1$ items *always* have lower support than itemsets with N items.
- D. Apriori uses a regression model to efficiently calculate the support of any itemset exploiting the downward closure property to find relevant inputs to the regression model.
- E. Don't know.

Solution 25. The correct answer is B.

The Apriori algorithm operates based on the Apriori principle, which states that if an itemset is frequent, then all of its subsets must also be frequent.

This is due to the downward closure property of support. By utilizing this principle, the algorithm prunes infrequent itemsets to efficiently search for frequent itemsets. It reduces the search space by eliminating itemsets that cannot be frequent based on the subsets' support levels.

Question 26. Which one of the following statements regarding cross-validation is *correct*?

- A. The size of the test set in 10 fold cross-validation is bigger than the size of the test set in 5 fold cross-validation.
- B. Leave-one-out cross-validation gives a poor estimate of the generalization error as only one observation is part of the test set at the time.
- C. The test error obtained for the model selected based on cross-validation is a biased estimate of the generalization error.**
- D. In K -fold cross-validation the trained models are more similar to each other compared to leave-one-out cross-validation because fewer data are used in each iteration for training.
- E. Don't know.

Solution 26.

- This is wrong. If N is the number of the data points, for 10-fold CV only $N/10$ are used for testing which is smaller than $N/5$.
- This is wrong. For a fixed dataset LOOCV gives the best estimation of the generalization error for this particular dataset (Setup I).
- Correct. The CV returns an estimate for generalization error which is, in general, optimistic i.e., below the true generalization error. This bias comes from the fact that we select always the smallest value, and since we use only a finite dataset our estimates typically distribute around the true generalization error.
- This is wrong. In LOOCV the data we use to train the model in each iteration are very correlated as they differ only in 2 points, so the learned models are very similar to each other. Instead, for K -fold CV, the correlation among datasets in each iteration is smaller, so the learned models exhibit more variability.

Question 27. Let $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denote the multivariate normal distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. In Figure 19 is given 1000 observations drawn from a density defined by a Gaussian Mixture Model

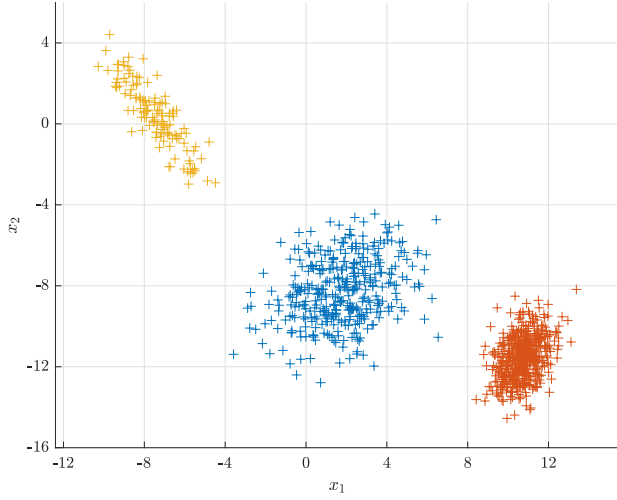


Figure 19: 1000 observations drawn from a Gaussian Mixture Model (GMM) with three clusters.

(GMM) with three clusters. Each observation is colored in terms of which cluster it came from in the Gaussian Mixture model.

Which one of the following GMM densities was used

to generate the data?

A.

$$p(\mathbf{x}) = \frac{3}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 1.7 \\ -8.4 \end{bmatrix}, \begin{bmatrix} 3.0 & 0.8 \\ 0.8 & 2.1 \end{bmatrix}\right) \\ + \frac{1}{2}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 10.7 \\ -11.5 \end{bmatrix}, \begin{bmatrix} 0.6 & 0.3 \\ 0.3 & 1.1 \end{bmatrix}\right) \\ + \frac{1}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -7.4 \\ 0.3 \end{bmatrix}, \begin{bmatrix} 1.3 & -1.5 \\ -1.5 & 2.3 \end{bmatrix}\right)$$

B.

$$p(\mathbf{x}) = \frac{1}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 1.7 \\ -8.4 \end{bmatrix}, \begin{bmatrix} 1.3 & -1.5 \\ -1.5 & 2.3 \end{bmatrix}\right) \\ + \frac{3}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 10.7 \\ -11.5 \end{bmatrix}, \begin{bmatrix} 0.6 & 0.3 \\ 0.3 & 1.1 \end{bmatrix}\right) \\ + \frac{1}{2}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -7.4 \\ 0.3 \end{bmatrix}, \begin{bmatrix} 3.0 & 0.8 \\ 0.8 & 2.1 \end{bmatrix}\right)$$

C.

$$p(\mathbf{x}) = \frac{1}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 1.7 \\ -8.4 \end{bmatrix}, \begin{bmatrix} 3.0 & 0.8 \\ 0.8 & 2.1 \end{bmatrix}\right) \\ + \frac{1}{2}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 10.7 \\ -11.5 \end{bmatrix}, \begin{bmatrix} 0.6 & 0.3 \\ 0.3 & 1.1 \end{bmatrix}\right) \\ + \frac{3}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -7.4 \\ 0.3 \end{bmatrix}, \begin{bmatrix} 1.3 & -1.5 \\ -1.5 & 2.3 \end{bmatrix}\right)$$

D.

$$p(\mathbf{x}) = \frac{3}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 1.7 \\ -8.4 \end{bmatrix}, \begin{bmatrix} 1.3 & -1.5 \\ -1.5 & 2.3 \end{bmatrix}\right) \\ + \frac{1}{2}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} 10.7 \\ -11.5 \end{bmatrix}, \begin{bmatrix} 3.0 & 0.8 \\ 0.8 & 2.1 \end{bmatrix}\right) \\ + \frac{1}{8}\mathcal{N}\left(\mathbf{x} \mid \begin{bmatrix} -7.4 \\ 0.3 \end{bmatrix}, \begin{bmatrix} 0.6 & 0.3 \\ 0.3 & 1.1 \end{bmatrix}\right)$$

E. Don't know.

Solution 27. The three components in the candidate GMM densities can be matched to the colored observations by their mean values. Then, by considering the basic properties of the covariance matrices, we can easily rule out all options except A. Alternatively, in Figure 20 is shown the densities for densities corresponding to option B (upper left), C (upper right) and D (bottom center).

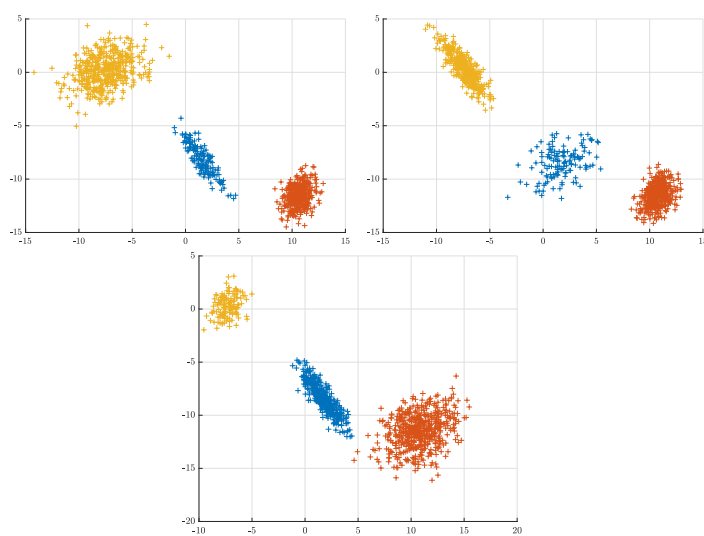


Figure 20: GMM mixtures corresponding to alternative options.